SOURAV TIWARI

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LinkedIn | GitHub | Portfolio

CAREER OBJECTIVE

B.Tech Computer Science student (2027) with hands-on experience building scalable, full-stack web applications using the MERN stack, REST API architecture, and relational/NoSQL database design, alongside AI/ML project work in Python. Skilled at translating business requirements into clean, well-documented code and delivering reliably within Agile sprint cycles. SIH 2025 Finalist and Microsoft Learn Student Ambassador.

TECHNICAL SKILLS

Languages:	Java, C, C++, Python, SQL
Development:	HTML, CSS, JavaScript, Node.js, Express.js, RESTful API Design, JWT Authentication
ML & AI:	Machine Learning, LLM Integration, Ollama, Prompt Engineering, AI Agents, Google Gemini AI
Databases:	MySQL, PostgreSQL, MongoDB, Database Design, Query Optimization
Tools & Platforms:	Git, GitHub, Postman, n8n, AWS, Google Cloud, Linux, Vercel, Render
Core Skills:	Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, SDLC, Agile/Scrum, Software Testing, Root Cause Analysis

INTERNSHIPS

Full Stack Development Intern — *Coding Arena* *Jul 2026 – Present*

- Developed responsive full-stack applications with React.js, Node.js, Express.js, and MongoDB, delivering RESTful APIs, secure authentication, and CRUD functionality for scalable software solutions.
- Collaborated with team members throughout the Software Development Life Cycle (SDLC), performed testing and debugging, maintained clean code using Git, and ensured timely delivery of high-quality features following Agile development practices.

Stack: *React.js, Node.js, Express.js, MongoDB, JavaScript, HTML5, CSS3, REST APIs, JWT Authentication, Git, GitHub, Postman, Vercel, Render*

ML and Backend Developer Intern — *Coding Arena* *Mar 2026 – Jun 2026*

- Architected 15+ REST API endpoints using Node.js and Express.js, reducing average response time by 25%.
- Engineered full-stack modules connecting React.js interfaces to Node.js services and designed backend database schema, accelerating delivery across 3 Agile product sprints.
- Developed and optimized machine learning models in Python, improving prediction accuracy by 20% across 5 classification pipelines while automating data workflows to save 8 hours per week.

Stack: *Node.js, Express.js, React.js, Python, Machine Learning, REST APIs, MongoDB, Agile*

PROJECTS

Hospital Management System — *Anveshana 2025*

- Analyzed client requirements and developed a responsive full-stack web app managing 100+ hospital records (patients, doctors, rooms) with real-time updates; documented system design, API specifications, and test cases, improving data access efficiency by ~30%.

Stack: *Node.js, Express.js, React, MongoDB, TypeScript, JWT Authentication, RESTful API*

AI/ML Crypto Primitive Identification in Firmware — *SIH 2025*

- Built an AI/ML pipeline to identify cryptographic primitives in firmware binaries; performed root cause analysis on false positives, documented performance requirements and validation scenarios, improving detection accuracy and reducing manual analysis effort by ~40%.
- Evaluated software-hardware behavior of firmware modules; prepared technical reports on system specifications, activities, and output for stakeholder review.

Stack: *Python, PyTorch, NumPy, NetworkX, Flask, HTML, CSS, JavaScript*

CertifiProof – AI-Powered Certificate Validation Platform — *Sageathon 2026*

- Built an AI-powered certificate authenticity validator using EfficientNet-B4 (6-channel RGB+ELA model) with ELA forensic analysis to detect tampered image regions; achieved automated fraud detection with GradCAM visual explanations of model verdicts.
- Integrated Tesseract OCR for field extraction, fuzzy-matched institutions against 50 Indian universities (NAAC), and deployed phi3/llama3 via Ollama for LLM-based forensic reasoning; delivered downloadable PDF reports and email delivery via Gmail SMTP with JWT authentication.

Stack: *Python, EfficientNet-B4, Tesseract OCR, Ollama (phi3/llama3), GradCAM, JWT, Flask, Gmail SMTP*

EDUCATION

Bachelor of Technology — Computer Science & Information Technology
Sagar Institute of Research and Technology, Bhopal

CPI: 8.13/10.00 | Coursework: DSA, OOP, DBMS, Operating Systems *2023 – 2027*

Intermediate (PCM), Government Model Co-Education Senior Secondary School, Punjab / *88.4%* *2021 – 2023*
Matriculation, St. Pius Convent Senior Secondary School, Punjab / *96.99%* *2019 – 2021*

CERTIFICATIONS

- NPTEL Introduction to Machine Learning — NPTEL | Scored 57% | 2026
- Google Cloud Arcade — Google | Completed 10+ cloud labs, multiple GCP skill badges | June 2025
- NPTEL Programming in Java — NPTEL | Scored 68% | 2025

ROLES & RESPONSIBILITIES

- Microsoft Learn Student Ambassador | Microsoft | April 2025 – Present — Engaged 100+ learners in web development and cloud topics across 5+ workshops.

EXTRA & CO-CURRICULAR ACTIVITIES

- Publications: Research paper published and presented at an International Conference (IJAR, ISSN: 2455-6211), June 2024.
- SIH 2025 Finalist — Selected among top teams nationally out of 10,000+ Smart India Hackathon entries.
- YUKTI 2025 & Anveshana 2025 — Reached the semifinals in two national-level innovation competitions.
- SolveExpo 2026 — Awarded 3rd Runner-Up for presenting an innovative web-based project to 50+ judges.